

Lifelong Multi-Agent Path Finding for Online Pickup and Delivery Tasks*

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ABSTRACT

The multi-agent path-finding (MAPF) problem has recently received a lot of attention. However, it does not capture important characteristics of many real-world domains, such as automated warehouses, where agents are constantly engaged with new tasks. In this paper, we therefore study a lifelong version of the MAPF problem, called the multi-agent pickup and delivery (MAPD) problem. In the MAPD problem, agents have to attend to a stream of delivery tasks in an online setting. One agent has to be assigned to each delivery task. This agent has to first move to a given pickup location and then to a given delivery location while avoiding collisions with other agents. We present two decoupled MAPD algorithms, Token Passing (TP) and Token Passing with Task Swaps (TPTS). Theoretically, we show that they solve all well-formed MAPD instances, a realistic subclass of MAPD instances. Experimentally, we compare them against a centralized strawman MAPD algorithm without this guarantee in a simulated warehouse system. TP can easily be extended to a fully distributed MAPD algorithm and is the best choice when real-time computation is of primary concern since it remains efficient for MAPD instances with hundreds of agents and tasks. TPTS requires limited communication among agents and balances well between TP and the centralized MAPD algorithm.

Keywords

agent coordination; multi-agent path finding; path planning; pickup and delivery tasks; task assignment

1. INTRODUCTION

Many real-world applications of multi-agent systems require agents to operate in known common environments.

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The agents are constantly engaged with new tasks and have to navigate between locations where the tasks need to be executed. Examples include aircraft-towing vehicles [18], warehouse robots [31], office robots [28], and game characters in video games [21]. In the near future, for instance, aircraft-towing vehicles might navigate autonomously to aircraft and tow them from the runways to their gates so as to reduce pollution, energy consumption, congestion, and human workload. Today, warehouse robots already navigate autonomously to inventory pods and move them from their storage locations to packing stations.

Past research efforts have concentrated mostly on a “one-shot” version of this problem, called the multi-agent path-finding (MAPF) problem, which has been studied in artificial intelligence, robotics, and operations research. In the MAPF problem, each agent has to move from its current location to its destination while avoiding collisions with other agents in a known common environment. The number of agents is the same as the number of destinations, and the MAPF task ends once all agents reach their destinations. Therefore, the MAPF problem does not capture important characteristics of many real-world domains, such as automated warehouses, where agents are constantly engaged with new tasks.

In this paper, we therefore study a “lifelong” version of the MAPF problem, called the multi-agent pickup and delivery (MAPD) problem. In the MAPD problem, agents have to attend to a stream of delivery tasks in a known common environment that is modeled as an undirected graph. Tasks can enter the system at any time and are modeled as exogenous events that are characterized by a pickup location and a delivery location each. An agent that is currently not executing any task can be assigned to an unexecuted task. In order to execute the task, the agent has to first move from its current location to the pickup location and then to the delivery location of the task while avoiding collisions with other agents. We first formalize the MAPD problem and then present two decoupled MAPD algorithms, Token Passing (TP) and Token Passing with Task Swaps (TPTS), both of which are based on existing MAPF algorithms. Theoretically, we show that they solve all well-formed MAPD instances [2], a realistic subclass of MAPD instances. Experimentally, we compare them against a centralized strawman MAPD algorithm without this guarantee in a simulated warehouse system.

2. BACKGROUND AND RELATED WORK

The MAPD problem requires both the assignment of agents to tasks in an online and lifelong setting and the planning of collision-free paths. In a lifelong setting, agents have to attend to a stream of tasks. Therefore, agents cannot rest in their destinations after they finish executing tasks. In an online setting, tasks can enter the system at any time. Therefore, assigning agents to tasks and path planning cannot be done in advance but rather need to be done during execution in real-time.

The decentralized assignment of agents to more than one task each has been studied before in isolation [26, 33, 17]. The decentralized planning of collision-free paths has also been studied before in isolation, including with reactive approaches [27] and prioritized approaches [6], but these approaches can result in deadlocks. The planning of collision-free paths has also been studied in the context of the MAPF problem, which is a one-shot (as opposed to a lifelong) version of the MAPD problem. It is NP-hard to solve optimally for minimizing flowtime (the sum of the number of timesteps required by all agents to reach their destinations and stop moving) and NP-hard to approximate within any constant factor less than 4/3 for minimizing makespan (the timestep when all agents have reached their destinations and stop moving) [16]. It can be solved via reductions to Boolean Satisfiability [25], Integer Linear Programming [32], and Answer Set Programming [5]. Optimal dedicated MAPF algorithms include Independence Detection with Operator Decomposition [22], Enhanced Partial Expansion A* [7], Increasing Cost Tree Search [20], M* [29], and Conflict-Based Search [19, 1, 3]. Suboptimal dedicated MAPF algorithms include Windowed-Hierarchical Cooperative A* [21, 23], Push and Swap/Rotate [12, 4], TASS [10], BIBOX [24], and MAPP [30]. The MAPF problem has recently been generalized to more clearly resemble real-world settings [8, 14, 9, 15, 13] but these versions are still one-shot.

3. PROBLEM DEFINITION

In this section, we first formalize the MAPD problem and then define well-formed MAPD instances.

3.1 MAPD Problem

An instance of the MAPD problem consists of m agents $A = \{a_1, a_2 \dots a_m\}$ and an undirected connected graph $G = (V, E)$ whose vertices V correspond to locations and whose edges E correspond to connections between locations that the agents can move along. Let $l_i(t) \in V$ denote the location of agent a_i in discrete timestep t . Agent a_i starts in its initial location $l_i(0)$. In each timestep t , the agent either stays in its current location $l_i(t)$ or moves to an adjacent location, that is, $l_i(t+1) = l_i(t)$ or $(l_i(t), l_i(t+1)) \in E$. Agents need to avoid collisions with each other: (1) Two agents cannot be in the same location in the same timestep, that is, for all agents a_i and $a_{i'}$ with $a_i \neq a_{i'}$ and timesteps t : $l_i(t) \neq l_{i'}(t)$; and (2) two agents cannot move along the same edge in opposite directions in the same timestep, that is, for all agents a_i and $a_{i'}$ with $a_i \neq a_{i'}$ and all timesteps t : $l_i(t) \neq l_{i'}(t+1)$ or $l_{i'}(t) \neq l_i(t+1)$. A path is a sequence of locations with associated timesteps, that is, a mapping from an interval of timesteps to locations. Two paths collide iff the two agents that move along them collide.

Consider a task set $\mathcal{T}$ that contains the set of unexecuted tasks. In each timestep, the system adds all new tasks

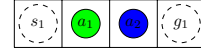


Figure 1: The figure shows a MAPD instance with two free agents a_1 and a_2 and one task τ_1 with pickup location s_1 and delivery location g_1 .

to the task set. Each task $\tau_j \in \mathcal{T}$ is characterized by a pickup location $s_j \in V$ and a delivery location $g_j \in V$. An agent is called free iff it is currently not executing any task. Otherwise, it is called occupied. A free agent can be assigned to any task $\tau_j \in \mathcal{T}$. It then has to move from its current location via the pickup location s_j of the task to the delivery location g_j of the task. (Any agent that had been assigned to this task previously no longer has this obligation.) When the agent reaches the pickup location, it starts to execute the task and the task is removed from $\mathcal{T}$. When it reaches the delivery location, it finishes executing the task, which implies that it becomes free again and is no longer assigned to the task. Note that any free agent can be assigned to any task in the task set. An agent can be assigned to a different task in the task set while it is still moving to the pickup location of the task it is currently assigned to but it has first to finish executing the task after it has reached its pickup location. These properties model delivery tasks, where agents can often be re-tasked before they have picked up a good but have to deliver it afterward.

The objective is to finish executing each task as quickly as possible. Consequently, the effectiveness of a MAPD algorithm is evaluated by the average number of timesteps, called service time, needed to finish executing each task after it was added to the task set. A MAPD algorithm solves a MAPD instance iff the resulting service time of all tasks is bounded.

3.2 Well-Formed MAPD Instances

Not every MAPD instance is solvable. Figure 1 shows an example with two free agents a_1 and a_2 where neither agent can finish executing task τ_1 with pickup location s_1 and delivery location g_1 . We now provide a sufficient condition that makes MAPD instances solvable, namely being well-formed [2]. The intuition is that agents should only be allowed to rest (that is, stay forever) in locations, called endpoints, where they cannot block other agents. For example, office workspaces are typically placed in office environments so as not to block routes. The set V_{ep} of endpoints of a MAPD instance contains all initial locations of agents, all pickup and delivery locations of tasks, and perhaps additional designated parking locations. Let V_{tsk} denote the set of all possible pickup and delivery locations of tasks, called the task endpoints. The set $V_{ep} \setminus V_{tsk}$ is called the set of non-task endpoints.

Definition 1. A MAPD instance is well-formed iff a) the number of tasks is finite, b) there are no fewer non-task endpoints than the number of agents, and c) for any two endpoints, there exists a path between them that traverses no other endpoints.

Well-formed MAPD instances (with at least one task) have at least $m+1$ endpoints. Figure 2 shows three MAPD instances. The MAPD instance on the left is well-formed. The MAPD instance in the center is not well-formed because there are two agents but only one non-task endpoint. The

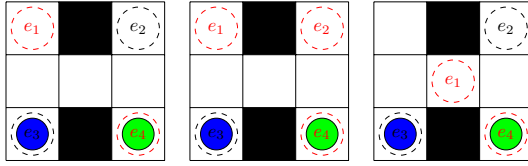


Figure 2: The figure shows three MAPD instances. Black cells are blocked. Blue and green circles are the initial locations of agents. Red dashed circles are task endpoints. Black dashed circles are non-task endpoints.

MAPD instance on the right is not well-formed because, for example, all paths between endpoints e_2 and e_3 traverse endpoint e_1 . We design two decoupled MAPD algorithms in the following that both solve all well-formed MAPD instances (even though they might not execute all tasks in case the number of tasks is infinite).

4. DECOUPLED MAPD ALGORITHMS

In this section, we present first a simple decoupled MAPD algorithm, called Token Passing (TP), and then an improved version, called Token Passing with Task Swaps (TPTS), that is more effective. Decoupled MAPD algorithms are those where each agent assigns itself to tasks and computes its own collision-free paths given some global information.

4.1 Token Passing (TP)

Token Passing (TP) is based on an idea similar to Cooperative A* [21], where agents plan their paths one after the other. Its task set contains all tasks that have no agents assigned to them. We describe a version of TP that uses token passing and can thus easily be extended to a fully distributed MAPD algorithm. The token is a synchronized shared block of memory that contains the current paths of all agents, task set, and agent assignments. All MAPD algorithms in this paper, including TP, always assume that an agent rests (that is, stays forever) in the last location of its path in the token when it reaches the end of it. Token passing has previously been used to develop COBRA [2], which is a MAPF-like algorithm that does not take into account that pickup or delivery locations of tasks can be occupied by agents not executing them and can thus result in deadlocks.

Algorithm 1 shows the pseudo-code of TP, where $loc(a_i)$ denotes the current location of agent a_i . Agent a_i finds all paths via A* searches in a state space whose states are pairs of locations and timesteps. A directed edge exists from state (l, t) to state $(l', t+1)$ iff $l = l'$ or $(l, l') \in E$. State (l, t) is removed from the state space iff a_i being in location l at timestep t results in it colliding with other agents that move along their paths in the token. Similarly, the edge from state (l, t) to state $(l', t+1)$ is removed from the state space iff a_i moving from location l to location l' at timestep t results in it colliding with other agents that move along their paths in the token. Since cost-minimal paths need to be found only to endpoints, the path costs from all locations to all endpoints are computed in a preprocessing phase and then used as h-values for all A* searches. TP works as follows: The system initializes the token with the trivial paths where all agents rest in their initial locations [Line 2]. In each

Algorithm 1: Token Passing (TP)

```

1 /* system executes now */;
2 Initialize token with the (trivial) path  $[loc(a_i)]$  for each agent  $a_i$ ;
3 while true do
4   Add all new tasks, if any, to the task set  $\mathcal{T}$ ;
5   while agent  $a_i$  exists that requests token do
6     /* system sends token to  $a_i$  -  $a_i$  executes now */;
7      $\mathcal{T}' \leftarrow \{\tau_j \in \mathcal{T} \mid \text{no other path in } token \text{ ends in } s_j \text{ or } g_j\}$ ;
8     if  $\mathcal{T}' \neq \emptyset$  then
9        $\tau \leftarrow \arg \min_{\tau_j \in \mathcal{T}'} h(loc(a_i), s_j)$ ;
10      Assign  $a_i$  to  $\tau$ ;
11      Remove  $\tau$  from  $\mathcal{T}$ ;
12      Update  $a_i$ 's path in token with  $Path1(a_i, \tau, token)$ ;
13    else if no task  $\tau_j \in \mathcal{T}$  exists with  $g_j = loc(a_i)$  then
14      Update  $a_i$ 's path in token with the path  $[loc(a_i)]$ ;
15    else
16      Update  $a_i$ 's path in token with  $Path2(a_i, token)$ ;
17    /*  $a_i$  returns token to system - system executes now */;
18  All agents move along their paths in token for one timestep;
19 /* system advances to the next timestep */;

```

timestep, the system adds all new tasks, if any, to the task set [Line 4]. Any agent that has reached the end of its path in the token requests the token once per timestep. (It turns out that one can easily drop the condition and let any free agent request the token once per timestep for both decoupled MAPD algorithms in this paper.) The system then sends the token to each agent that requests it, one after the other [Lines 5-6]. The agent with the token chooses a task from the task set such that no path of other agents in the token ends in the pickup or delivery location of the task [Line 7].

- If there is at least one such task, then the agent assigns itself to the one with the smallest h-value from its current location to the pickup location of the task and removes this task from the task set [Lines 9-11]. The agent then calls function $Path1$ to update its path in the token with a cost-minimal path that a) moves from its current location via the pickup location of the task to the delivery location of the task and b) does not collide with the paths of other agents stored in the token [Line 12].
- If there is no such task, then the agent does not assign itself to a task in the current timestep. If the agent is not in the delivery location of a task in the task set, then it updates its path in the token with the trivial path where it rests in its current location [Line 14]. Otherwise, to avoid deadlocks, it calls function $Path2$ to update its path in the token with a cost-minimal path that a) moves from its current location to an endpoint such that the delivery locations of all tasks in the task set are different from the chosen endpoint and no path of other agents in the token ends in the chosen endpoint and b) does not collide with the paths of other agents stored in the token [Line 16].

Finally, the agent returns the token to the system and moves along its path in the token [Lines 17-18].

We now prove that the agent is always able to find a path because it finds a path only when it is at an endpoint and thus has to find only a path from an endpoint to an endpoint.

Property 1. Function $Path1$ returns a path successfully for well-formed MAPD instances.

Proof. We construct a path from the current location $loc(a_i)$ of agent a_i (which is an endpoint) via the pickup location s_j of task τ_j to the delivery location g_j of task τ_j that does not collide with the paths of other agents stored in the token. Due to Definition 1, there exists a path from $loc(a_i)$ via s_j to g_j that traverses no other endpoints. All paths stored in $token$ end in endpoints that are different from $loc(a_i)$, s_j , and g_j . Thus, this path does not collide with the paths of the other agents if a_i moves along it after all other agents have moved along their paths. $\square$

Property 2. Function Path2 returns a path successfully for well-formed MAPD instances.

Proof. Due to Definition 1, there exist at least m non-task endpoints and thus at least one non-task endpoint such that no path of agents other than agent a_i in the token ends in the non-task endpoint. Of course, the delivery locations of all tasks in the task set are different from the non-task endpoint as well. We construct a path from the current location $loc(a_i)$ of agent a_i (which is an endpoint) to the chosen endpoint that does not collide with the paths of other agents stored in the token. Due to Definition 1, there exists a path from $loc(a_i)$ to the chosen endpoint that traverses no other endpoints. All paths stored in $token$ end in endpoints that are different from $loc(a_i)$ and the chosen endpoint. Thus, this path does not collide with the paths of the other agents if a_i moves along it after all other agents have moved along their paths. $\square$

Theorem 3. All well-formed MAPD instances are solvable, and TP solves them.

Proof. We show that each task is eventually assigned some agent and executed by it. Each agent requests the token after a bounded number of timesteps, and no agent rests in the delivery location of a task in the task set due to Line 16. Thus, the condition on Line 8 becomes eventually satisfied and some agent assigns itself to some task on Line 10. The agent is then able to execute it due to Properties 1 and 2. $\square$

4.2 Token Passing with Task Swaps (TPTS)

TP is simple but can be made more effective. Token Passing with Task Swaps (TPTS) is similar to TP except that its task set now contains all unexecuted tasks, rather than only tasks that have no agents assigned. This means that an agent with the token can assign itself not only to a task that has no agent assigned but also to a task that is already assigned another agent as long as that agent is still moving to the pickup location of the task. This might be beneficial when the former agent can move to the pickup location of the task in fewer timesteps than the latter agent. The latter agent is then no longer assigned to the task and no longer needs to execute it. The former agent therefore sends the token to the latter agent so that the latter agent can try to assign itself to a new task.

Algorithm 2 shows the pseudo-code of TPTS. It uses the same main loop [Lines 3-10] and the same functions Path1 and Path2 as TP. Agent a_i with the token executes function GetTask [Line 7], where it tries to assign itself to a task in the task set $\mathcal{T}$ and find a path to an endpoint. The call of function GetTask returns success (true) if agent a_i finds a path to an endpoint and failure (false) otherwise.

Algorithm 2: Token Passing with Task Swaps (TPTS)

```

1  /* system executes now */;
2  Initialize  $token$  with the (trivial) path  $[loc(a_i)]$  for each agent  $a_i$ ;
3  while true do
4    Add all new tasks, if any, to the task set  $\mathcal{T}$ ;
5    while agent  $a_i$  exists that requests  $token$  do
6      /* system sends  $token$  to  $a_i$  -  $a_i$  executes now */;
7      GetTask( $a_i$ ,  $token$ );
8      /*  $a_i$  returns  $token$  to system - system executes now */;
9    All agents move along their paths in  $token$  for one timestep
    and remove tasks from  $\mathcal{T}$  when they start to execute them;
10   /* system advances to the next timestep */;

11  Function GetTask( $a_i$ ,  $token$ )
12    $\mathcal{T}' \leftarrow \{\tau_j \in \mathcal{T} \mid \text{no other path in } token \text{ ends in } s_j \text{ or } g_j\}$ ;
13   while  $\mathcal{T}' \neq \emptyset$  do
14      $\tau \leftarrow \arg \min_{\tau_j \in \mathcal{T}'} h(loc(a_i), s_j)$ ;
15     Remove  $\tau$  from  $\mathcal{T}'$ ;
16     if no agent is assigned to  $\tau$  then
17       Assign  $a_i$  to  $\tau$ ;
18       Update  $a_i$ 's path in  $token$  with Path1( $a_i$ ,  $\tau$ ,  $token$ );
19       return true;
20     else
21       Remember  $token$ , task set, and agent assignments;
22        $a_{i'} \leftarrow$  agent that is assigned to  $\tau$ ;
23       Unassign  $a_{i'}$  from  $\tau$  and assign  $a_i$  to  $\tau$ ;
24       Remove  $a_{i'}$ 's path from  $token$ ;
25       Path1( $a_i$ ,  $\tau$ ,  $token$ );
26       Compare when  $a_i$  reaches  $s_j$  on its path in  $token$  to
       when  $a_{i'}$  reaches  $s_j$  on its path in  $token$ ;
27       if  $a_i$  reaches  $s_j$  earlier than  $a_{i'}$  then
28         /*  $a_i$  sends  $token$  to  $a_{i'}$  -  $a_{i'}$  executes now */;
29         success  $\leftarrow$  GetTask( $a_{i'}$ ,  $token$ );
30         /*  $a_{i'}$  returns  $token$  to  $a_i$  -  $a_i$  executes now */;
31         if success then
32           return true;
33       Restore  $token$ , task set, and agent assignments;
34   if  $loc(a_i)$  is not an endpoint then
35     Update  $a_i$ 's path in  $token$  with Path2( $a_i$ ,  $token$ );
36     if path was found then
37       return true;
38   else
39     if no task  $\tau_j \in \mathcal{T}$  exists with  $g_j = loc(a_i)$  then
40       Update  $a_i$ 's path in  $token$  with the path  $[loc(a_i)]$ ;
41     else
42       Update  $a_i$ 's path in  $token$  with Path2( $a_i$ ,  $token$ );
43     return true;
44   return false;
```

When executing function GetTask, agent a_i considers all tasks τ from the task set such that no path of other agents in the token ends in the pickup or delivery location of the task [Line 12], one after the other in order of increasing h -values from its current location to the pickup locations of the tasks [Lines 13-15]. If no agent is assigned to the task, then (as in TP) agent a_i assigns itself to the task, updates its path in the token with function Path1 and returns success [Lines 17-19]. Otherwise, agent a_i unassigns the agent $a_{i'}$ assigned to the task and assigns itself to the task [Line 23]. It removes the path of agent $a_{i'}$ from the token and updates its own path in the token with function Path1 [Lines 24-25]. If agent a_i reaches the pickup location of the task with fewer timesteps than agent $a_{i'}$, then it sends the token to agent $a_{i'}$, which executes function GetTask to try to assign itself to a new task and eventually returns the token to agent a_i [Lines 28-30]. If agent $a_{i'}$ returns success, then agent a_i returns success as well [Lines 31-32]. In all other cases, agent a_i reverses all changes to the paths in the token, task set, and

agent assignments and then considers the next task τ [Lines 33].

Once agent a_i has considered all tasks τ unsuccessfully, then it does not assign itself to a task in the current timestep. If it is not in an endpoint (which can happen only during a call of Function GetTask on Line 29), then it updates its path in the token with function Path2 to move to an endpoint [Line 35]. The call can fail since agent a_i is not at an endpoint. Agent a_i returns success or failure depending on whether it was able to find a path [Lines 37 and 44]. Otherwise, (as in TP) if the agent is not in the delivery location of a task in the task set, then it updates its path in the token with the trivial path where it rests in its current location [Line 40]. Otherwise, to avoid deadlocks, it updates its path in the token with Function Path2 [Line 42]. In both cases, it returns success [Line 43].

Finally, the agent returns the token to the system and moves along its path in the token, removing the task that it is assigned to (if any) from the task set once it reaches the pickup location of the task (and thus starts to execute it) [Lines 8-9].

Property 4. Function GetTask returns successfully for well-formed MAPD instances when called on Line 7.

Proof. Function GetTask returns in finite time (and the number of times an agent can unassign another agent from any task is bounded during its execution) because a) the number of tasks in the task set is finite; b) an agent can unassign another agent from a task only if it reaches the pickup location of the task with fewer timesteps than the other agent; c) a task that has some agent assigned always continues to have some agent assigned until it has been executed; and d) the functions Path1 and Path2 return in finite time. We show that an agent that executes function GetTask on Line 7 finds a path to an endpoint. The agent is always able to find paths with functions Path1 and Path 2 on all lines but Line 35 because it is then at an endpoint and thus has to find a path from an endpoint to an endpoint. The proofs are similar to those of Properties 1 and 2. However, the agent is not guaranteed to find a path with function Path2 on Line 35 because it is then not at an endpoint and thus has to find a path from a non-endpoint to an endpoint. Since the agent is at an endpoint during the call of function GetTask on Line 7, it does not execute Line 35, finds a path, and returns success. $\square$

Theorem 5. TPTS solves all well-formed MAPD instances.

Proof. The proof is similar to the one of Theorem 3 but uses Property 4. $\square$

TPTS is often more effective than TP but Figure 3 shows that this is not guaranteed. The figure shows a MAPD instance with two agents a_1 and a_2 and two tasks τ_1 and τ_2 . The pickup location is the same as the delivery location for each task. Assume that both a_1 and a_2 request the token and the system sends it to a_1 first. a_1 assigns itself to τ_1 . Figure 3 (left) shows the path of a_1 . The system then sends the token to a_2 next. In TP, a_2 assigns itself to τ_2 . Figure 3 (center) shows the paths of a_1 and a_2 . The resulting service time is two. In TPTS, however, a_2 assigns itself to τ_1 because it can reach the pickup location of τ_1 with fewer timesteps than a_1 . In return, a_1 assigns itself

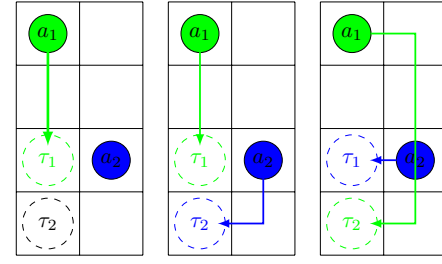


Figure 3: The figure shows a MAPF instance. The pickup location is the same as the delivery location for both two tasks. Blue and green circles are the initial locations of agents. Dashed circles are the pickup/delivery locations.

to τ_2 . Figure 3 (right) shows the paths of a_1 and a_2 . The service time is three (the average of five and one).

5. CENTRALIZED ALGORITHM

In this section, we develop a centralized strawman MAPD algorithm, CENTRAL, to evaluate how effective our decoupled MAPD algorithms are. We want CENTRAL to be reasonably efficient and effective but do not require that it is optimally effective or even solves all well-formed MAPD instances. The agents of a centralized MAPD algorithm can communicate more than the ones of TPTS, and the ones of TPTS communicate more than the ones of TP. Thus, we expect the MAPD algorithms to be in increasing order of effectiveness: TP, TPTS, and CENTRAL. For the same reason, we expect the MAPD algorithms to be in increasing order of efficiency: CENTRAL, TPTS, and TP.

In each timestep, CENTRAL first assigns endpoints to all agents and then solves the resulting MAPF instance to plan paths for all agents from their current locations to their assigned endpoints simultaneously. Finally, all agents move along their paths for one timestep and the procedure repeats.

Agent Assignment First, CENTRAL considers each agent, one after the other, that rests at the pickup location of an unexecuted task. If the delivery location of the task is currently not assigned to other agents, CENTRAL assigns the agent to the corresponding unexecuted task (if it is not assigned to the task already) and assigns the delivery location of the task to the agent. The agent then starts to execute the task and thus becomes occupied. Then, CENTRAL assigns each free agent either the pickup location of an unexecuted task or some other endpoint as parking location. To make the resulting MAPF problem solvable, the endpoints assigned to all agents must be pairwise different. Agents are assigned pickup locations of unexecuted tasks in order to execute the tasks afterward. Thus, when CENTRAL assigns pickup locations of unexecuted tasks to agents, we want the delivery locations of these tasks to be different from the endpoints assigned to all agents (except for their own pickup locations) and from each other. CENTRAL achieves these constraints as follows:

First, CENTRAL greedily constructs a set of possible endpoints X for the free agents as follows: CENTRAL greedily constructs a subset $\mathcal{T}'$ of unexecuted tasks, starting with the empty set, by checking for each unexecuted task, one after the other, whether its pickup and delivery locations are different from the delivery locations of all executed tasks

and the pickup and delivery locations of all unexecuted tasks already added to $\mathcal{T}'$ and, if so, adds it to $\mathcal{T}'$. CENTRAL then sets X to the pickup locations of all tasks in $\mathcal{T}'$. If the number of free agents is larger than $|X|$, then CENTRAL needs to add endpoints to X as parking locations for some free agents. Since it is not known a priori which free agents these parking locations will be assigned to, there should be one good parking location available for each free agent, which is possible due to Definition 1. CENTRAL thus greedily determines a good parking location for each free agent a_i , one after the other, as the endpoint e that minimizes the cost $c(a_i, e)$ (“is closest to the agent”) among all endpoints that are different from the delivery locations of all executed tasks, the pickup and delivery locations of all tasks in $\mathcal{T}'$, and the parking locations already determined, where $c(a_i, e)$ is the cost of a cost-minimal path that moves from the current location of free agent a_i to endpoint e . It then adds this endpoint to X .

Second, CENTRAL assigns each free agent an endpoint in X to satisfy all constraints. It uses the Hungarian Method [11] for this purpose with the modified costs $c'(a_i, e)$ for each pair of free agent a_i and endpoint e , where c is the number of free agents, C is a sufficiently large constant (for example, the maximum over all costs $c(a_i, e)$ plus one), and $c'(a_i, e) = c \cdot C \cdot c(a_i, e)$ if e is a pickup location of a task in $\mathcal{T}'$ and $c'(a_i, e) = c \cdot C^2 + c(a_i, e)$ if e is a parking location. The modified costs have two desirable properties: a) The modified cost of assigning a pickup location to a free agent is always smaller than the modified cost of assigning a parking location to the same agent. Therefore, assigning pickup locations is more important than assigning rest locations. b) Assigning a closer pickup location to a single free agent that is assigned a pickup location reduces the total modified cost more than assigning closer parking locations to all free agents that are assigned rest locations. Therefore, assigning closer pickup locations is more important than assigning closer parking locations.

Path Planning CENTRAL uses the optimally effective MAPF algorithm Conflict-Based Search [19] to plan collision-free paths for all agents from their current locations to their assigned endpoints simultaneously. These paths minimize the sum of the number of timesteps required by all agents to reach their assigned endpoints and stop moving. We noticed that CENTRAL becomes significantly more efficient if it plans paths in two stages: First, it plans paths for all agents that become occupied in the current timestep to their assigned endpoints (using the approach described above but treating the most recently calculated paths of all other agents as spatio-temporal obstacles). Then, it plans paths for all free agents to their assigned endpoints (again using the approach described above but treating the most recently calculated paths of all other agents as spatio-temporal obstacles). In general, two smaller MAPF instances can be solved much faster than their union due to the NP-hardness of the problem. Also, CENTRAL can then determine a more informed cost $c(a_i, e)$ as the cost of a cost-minimal path that a) moves from the current location of agent a_i to endpoint e and b) does not collide with the paths of the occupied agents (as described for TP).

Property 6. Path planning for all agents that became occupied in the current timestep returns paths successfully for well-formed MAPD instances.

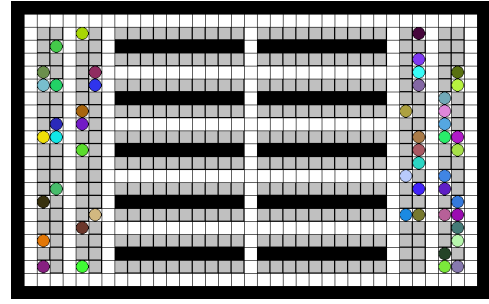


Figure 4: The figure shows a 21×35 4-neighbor grid that represents the layout of a small simulated warehouse environment with 50 agents. Black cells are blocked. Gray cells are task endpoints. Colored circles are the initial locations of agents.

Proof. We construct paths for all agents that became occupied in the current timestep from their current locations to their assigned endpoints that do not collide with the most recently calculated paths of all other agents: Assume that all other agents move along their most recently calculated paths. When all of them have reached the ends of their paths, move all agents that became occupied one after the other to their assigned endpoints, which is possible due to Definition 1 since their current locations are endpoints and their assigned endpoints are different from the endpoints that all other agents now occupy. $\square$

Property 7. Path planning for all free agents returns paths successfully for well-formed MAPD instances.

Proof. We construct paths for all free agents from their current locations to their assigned endpoints that do not collide with the most recently calculated paths of all other agents: Assume that all agents move along their most recently calculated paths. When all of them have reached the ends of their paths, move all free agents one after the other to their assigned endpoints, which is possible due to Definition 1 since the locations that they now occupy are endpoints. Directly before an agent moves to its assigned endpoint, check whether this endpoint is blocked by another agent. If so, move this other agent to an unoccupied endpoint first. Such an endpoint exists since there are at least $m + 1$ endpoints for m agents due to Definition 1. $\square$

6. EXPERIMENTAL EVALUATION

In this section, we describe our experimental results on a 2.50 GHz Intel Core i5-2450M laptop with 6 GB RAM. We ran TP, TPTS, and CENTRAL in the small simulated warehouse environment shown in Figure 4. We generated a sequence of 500 delivery tasks by randomly choosing their pickup and delivery locations from all task endpoints. The initial locations of the agents are the only non-task endpoints. We used 6 different task frequencies (numbers of tasks that are added (in order) from the sequence to the task set in each timestep): 0.2 (one task every 5 timesteps), 0.5, 1, 2, 5, and 10. For each task frequency, we used 5 different numbers of agents: 10, 20, 30, 40, and 50. Table 1 reports the makespans, the service times, the runtimes per timestep (in ms), the ratios of the service times of TPTS and TP, and the ratios of the service times of CENTRAL

Table 1: The table shows the experimental results for TP, TPTS, and CENTRAL in the small simulated warehouse environment.

task frequency	agents	TP				TPTS				CENTRAL			
		make-span	service time	run-time	ratio	make-span	service time	run-time	ratio	make-span	service time	run-time	ratio
0.2	10	2.532	38.54	0.13	2.532	29.33	1.86	0.76	2.513	27.78	92.69	0.72	
	20	2.540	39.77	0.26	2.520	25.36	9.82	0.64	2.513	24.37	493.83	0.61	
	30	2.546	38.71	0.25	2.527	23.88	21.57	0.62	2.513	23.10	1,225.62	0.60	
	40	2.540	38.88	0.24	2.524	23.50	27.49	0.60	2.511	22.48	2,246.66	0.58	
	50	2.540	40.03	0.32	2.524	23.11	47.33	0.58	2.511	21.82	3,426.01	0.54	
0.5	10	1.309	132.79	0.24	1.274	131.15	0.33	0.99	1.242	116.37	113.00	0.88	
	20	1.094	42.69	1.16	1.038	30.74	12.39	0.72	1.031	28.05	300.01	0.66	
	30	1.069	43.97	1.51	1.035	27.14	34.04	0.62	1.034	25.36	916.37	0.58	
	40	1.090	43.01	0.70	1.038	25.98	19.85	0.60	1.034	24.26	1,697.99	0.56	
	50	1.083	43.66	1.36	1.036	25.22	71.48	0.58	1.031	23.83	2,920.72	0.55	
1	10	1.198	311.78	0.20	1.182	301.03	0.37	0.97	1.143	285.67	141.32	0.92	
	20	757	95.98	1.03	706	88.25	2.80	0.92	673	74.79	279.52	0.78	
	30	607	53.80	2.81	561	42.84	8.45	0.80	557	30.26	446.38	0.56	
	40	624	48.80	2.14	563	31.99	39.54	0.66	556	28.27	1,159.76	0.58	
	50	597	49.14	3.76	554	30.27	128.13	0.62	552	26.55	2,197.82	0.54	
2	10	1.167	407.62	0.19	1.168	407.24	0.37	1.00	1.121	386.81	143.30	0.95	
	20	683	190.76	0.96	667	181.03	2.38	0.95	628	163.79	406.88	0.86	
	30	529	114.39	2.31	496	102.69	8.39	0.90	466	88.45	589.13	0.77	
	40	464	95.32	3.43	425	72.69	7.32	0.76	385	58.12	837.21	0.61	
	50	432	75.63	6.28	383	58.06	126.47	0.77	320	39.25	1,200.10	0.52	
5	10	1.162	473.78	0.20	1.165	473.18	0.41	1.00	1.105	452.50	126.19	0.96	
	20	655	247.08	1.02	645	238.02	1.68	0.96	594	224.70	350.13	0.91	
	30	478	170.78	2.22	474	167.66	8.58	0.98	426	147.03	595.04	0.86	
	40	418	155.33	4.15	396	131.36	12.31	0.85	334	108.39	864.56	0.70	
	50	395	124.59	5.92	343	104.86	59.64	0.84	295	86.22	1,388.30	0.69	
10	10	1.163	495.93	0.22	1.172	505.26	0.40	1.02	1.090	472.56	125.55	0.95	
	20	643	275.24	1.09	645	258.36	1.87	0.94	607	248.74	379.53	0.90	
	30	526	192.01	1.98	491	198.30	10.82	1.03	414	164.41	593.89	0.86	
	40	407	154.63	1.65	389	152.49	12.62	0.99	341	128.29	899.81	0.83	
	50	333	131.42	5.62	319	126.96	25.32	0.97	277	105.11	1,376.51	0.80	

and TP. The measures for a task frequency of 10 tasks per timestep are reasonably representative of the case where all tasks are added in the beginning of the operation since the tasks are added over the first 50 timesteps only.

Makespans and Service Times The MAPD algorithms in increasing order of their makespans and service times tend to be: CENTRAL, TPTS, and TP. For example, the service time of TPTS (and CENTRAL) is up to about 42 percent (and 48 percent, respectively) smaller than the one of TP for some experimental runs. The makespans tend to be large for low task frequencies and small for high task frequencies because the number of tasks is constant and thus more time steps are needed to add all tasks for low task frequencies. On the other hand, the service times tend to be small for low task frequencies and high for high task frequencies because the agents tend to be able to attend to tasks fast if the number of tasks in the system is small. The makespans and service times tend to be large for small numbers of agents and small for large numbers of agents because the agents tend to be able to attend to tasks fast if the number of agents is large (although congestion increases). The makespans and service times for a task frequency of 0.2 tasks per timestep are about the same for all numbers of agents because 10 agents already attend to all tasks as fast as the MAPD algorithms allow. The makespans are similar for all MAPD algorithms and all numbers of agents for the task frequency of 0.2 tasks per timestep because 10 agents already execute tasks faster than they are added. The makespans then depend largely on how fast the agents execute the last few tasks. On the other hand, the makespans of MAPD algorithms increase substantially when tasks pile up because the agents execute them more slowly than they are added. This allows us to estimate the smallest number of agents needed for a lifelong operation as a function of the task frequency and MAPD algorithm. For example, the makespan of TPTS increases substantially when the number of agents are reduced from 20 to 10 for a task frequency of 1 task per timestep. Thus, one needs between about 10 and 20 agents for a lifelong operation with TPTS.

Runtimes per Timestep The MAPF algorithms in increasing order of their runtimes per timestep tend to be: TP, TPTS, and CENTRAL. For example, the runtime of TPTS (and CENTRAL) is two orders of magnitude larger than the runtime of TP (and TPTS, respectively) for some experimental runs. The runtimes of TP are less than 10 milliseconds, the runtimes of TPTS are less than 200 milliseconds, and the runtimes of CENTRAL are less than 4,000 milliseconds in all experimental runs. We consider runtimes below one second to allow for real-time lifelong operation. The runtimes tend not to be correlated with the task frequencies. They tend to be small for small numbers of agents and large for large numbers of agents because all agents need to perform computations, which are not run in parallel in our experiments.

Number of Executed Tasks The service times vary over time since only very few tasks are available in the first and last time steps. The steady state is in between these two extremes. Figure 5 therefore visualizes the number of tasks added and executed by 50 agents during the 100-timestep window $[t - 99, t]$ for all MAPD algorithms as a function of the timestep t . For low task frequencies, the numbers of tasks added match the numbers of tasks executed closely for all MAPD algorithms. Differences between them arise for higher task frequencies. For example, for the task frequency of 2 tasks per timestep, the number of tasks executed by CENTRAL increases faster and reaches a higher level than the numbers of tasks executed by TP and TPTS. The 100-timestep window $[150, 249]$ at time step $t = 249$ is a close approximation of the steady state since all tasks are added at a steady rate during the first 250 timesteps. CENTRAL executes more tasks during this 100-timestep window than TP and TPTS and thus has a smaller service time. However, the numbers of tasks executed are smaller for all MAPD algorithms than the number of tasks added, and tasks thus pile up for all of them in the steady state.

Scalability To evaluate how the MAPD algorithms scale in the size of the environment, we ran TP, TPTS, and CENTRAL in the large simulated warehouse environment shown in Figure 6. We generated a sequence of 1,000 delivery tasks by randomly choosing their pickup and delivery locations from all task endpoints. The initial locations of the agents are the only non-task endpoints. We used a task frequency of 50 tasks per timestep and 100, 200, 300, 400, and 500 agents. TPTS and CENTRAL did not allow for real-time lifelong operation for large numbers of agents. Figure 7 therefore reports, for TP only, the service times and the runtimes per timestep (in ms) in the table as well as the numbers of tasks added and executed during a 100-timestep window for different numbers of agents in the charts. The runtime of TP is smaller than 500 milliseconds for 200 agents, allowing for real-time lifelong operation.

7. CONCLUSIONS

In this paper, we studied a lifelong version of the multi-agent path finding (MAPF) problem, called the multi-agent pickup and delivery (MAPD) problem, to capture important characteristics of many real-world domains. In the MAPD problem, agents have to attend to a stream of delivery tasks in an online setting by first moving to the pickup locations of the tasks and then to the delivery locations of the tasks while avoiding collisions with other agents. We presented two decoupled MAPD algorithms, Token Passing (TP) and

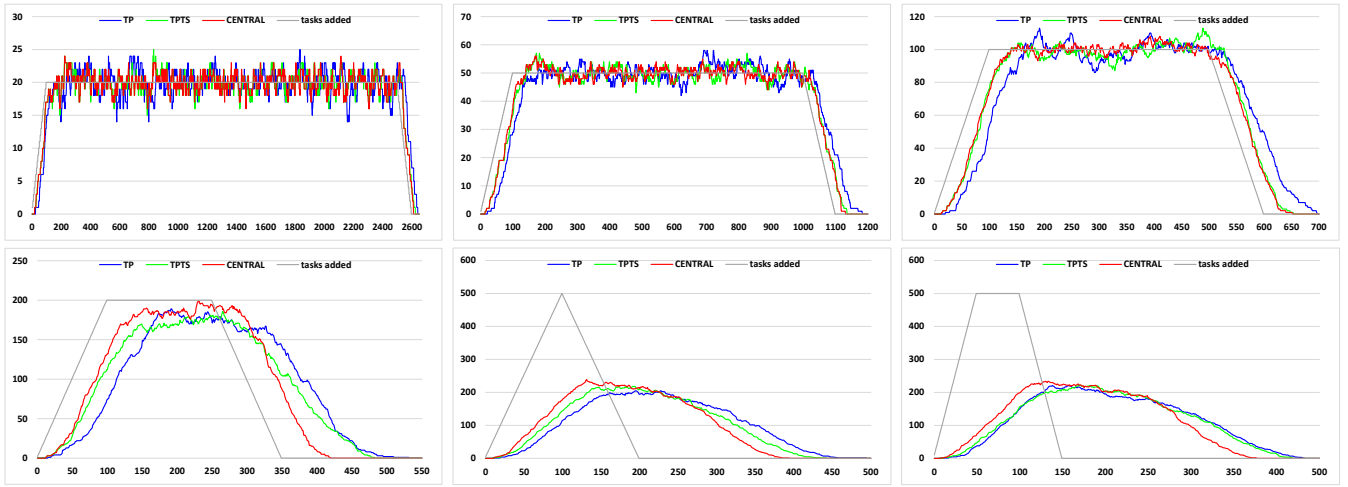


Figure 5: The charts show the number of tasks added (grey) and executed by 50 agents during the 100-timestep window $[t - 99, t]$ for TP, TPTS, and CENTRAL as a function of the timestep t for task frequencies 0.2, 0.5, 1, 2, 5, and 10 (from left to right and top to bottom)

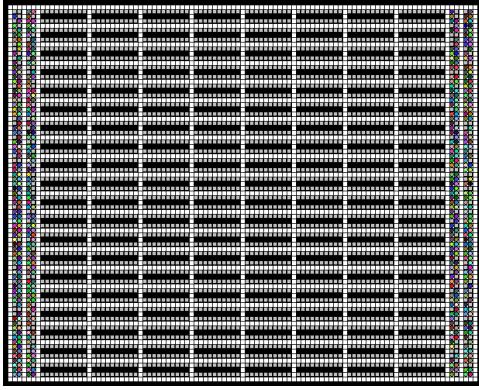


Figure 6: The figure shows a 81×81 4-neighbor grid that represents the layout of a large simulated warehouse environment with 500 agents.

Token Passing with Task Swaps (TPTS). Theoretically, we showed that both MAPD algorithms solve all well-formed MAPD instances. Experimentally, we compared them against the centralized strawman MAPD algorithm CENTRAL without this guarantee in a simulated warehouse system. The MAPD algorithms in increasing order of their makespans and service times tend to be: CENTRAL, TPTS, and TP. The MAPF algorithms in increasing order of their runtimes per timestep tend to be: TP, TPTS, and CENTRAL. TP can easily be extended to a fully distributed MAPD algorithm and is the best choice when real-time computation is of primary concern since it remains efficient for MAPD instances with hundreds of agents and tasks. TPTS requires limited communication among agents and balances well between TP and CENTRAL.

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agents	100	200	300	400	500
service time	463.25	330.19	301.97	289.08	284.24
runtime	90.83	538.22	1,854.44	3,881.11	6,121.06

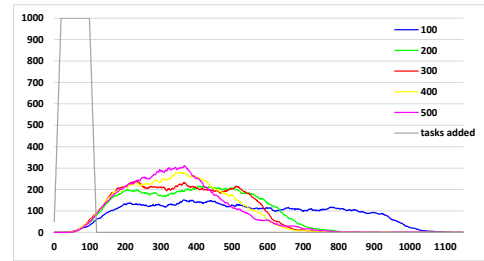


Figure 7: The figure shows the experimental results for TP in the large simulated warehouse environment.

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